

Estimating Volumes of Square and Rectangular Pyramids

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To estimate the volume, round the dimensions to convenient numbers before computing.

Estimation

Estimation is useful when exact measurements are unnecessary or when making quick calculations.

Examples

Example 1

A square pyramid has a base edge of 8.2 cm and a height of 11.6 cm. Estimate its volume.

Example 2

A square pyramid with a base edge of 12.4 m and a height of 9.7 m. Estimate its volume.

Example 3

A rectangular pyramid with a base of 15.3 m by 10.6 m and a height of 12.2 m. Estimate its volume.

Practice Exercises

Practice 1

Estimate the volume.

A square pyramid has a base edge of 8.2 cm and a height of 11.6 cm.

Practice 2

Estimate the volume.

A square pyramid with a base edge of 12.4 m and a height of 9.7 m.

Practice 3

Estimate the volume.

A rectangular pyramid with a base of 15.3 m by 10.6 m and a height of 12.2 m.

Practice 4

Estimate the volume.

A square pyramid has a base edge of 19.6 cm and a height of 14.8 cm.

Practice 5

Estimate the volume.

A rectangular pyramid with dimensions 13.5 cm, 8.2 cm, and height 16.9 cm.

Practice 6

Estimate the volume.

A square pyramid has a base edge of 24.7 m and a height of 18.3 m.

Practice 7

Estimate the volume.

A rectangular pyramid whose base is 18.8 cm by 11.4 cm and height is 15.6 cm.

Practice 8

Estimate the volume.

A square pyramid has a base edge of 31.2 cm and a height of 20.5 cm.

Activities

Activity 1

Estimate the volume.

A square pyramid has a base edge of 9.4 cm and a height of 12.3 cm.

Activity 2

Estimate the volume.

A square pyramid with a base edge of 13.8 m and a height of 10.5 m.

Activity 3

Estimate the volume.

A rectangular pyramid has a base measuring 8.6 cm by 6.4 cm and a height of 11.8 cm.

Activity 4

Estimate the volume.

A rectangular pyramid with a base of 16.9 m by 11.5 m and a height of 13.6 m.

Activity 5

Estimate the volume.

A square pyramid has a base edge of 21.1 cm and a height of 15.9 cm.

Activity 6

Estimate the volume.

A rectangular pyramid with dimensions 14.8 cm, 9.7 cm, and height 17.4 cm.

Activity 7

Estimate the volume.

A square pyramid has a base edge of 26.3 m and a height of 19.5 m.

Activity 8

Estimate the volume.

A rectangular pyramid whose base is 20.4 cm by 12.8 cm and height is 16.7 cm.

Activity 9

Estimate the volume.

A square pyramid has a base edge of 32.8 cm and a height of 21.9 cm.

Activity 10

Estimate the volume.

A rectangular pyramid with a base measuring 24.3 m by 18.6 m and a height of 26.8 m.

Thank you!